

Answer Key — Size, Atmosphere and Magnetic Field

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1** (b) **gravity** — enough gravity (from the right size) holds the atmosphere.
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- Q2** (b) **ultraviolet (UV) rays** — ozone absorbs harmful UV rays.
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- Q3** (b) **molten iron moving in the core** — moving molten iron in the core generates the field.
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- Q4** (a) **cosmic rays and solar wind** — it deflects cosmic rays and the solar wind.
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- Q5** (b) **False** — ozone is a different molecule — three oxygen atoms.
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- Q6** **ozone** — the ozone layer blocks UV.
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- Q7** **wind** — these particles are the solar wind.
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Short answer

- Q8** It gives just enough gravity to hold an atmosphere — too small and gases escape, too large and gravity could be crushing.
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- Q9** It blocks harmful ultraviolet (UV) rays from the Sun, protecting living cells.
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- Q10** It acts as a shield, deflecting harmful cosmic rays and the solar wind, protecting the atmosphere and life.
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Long / application

- Q11** (1) The right size gives enough gravity to hold an atmosphere. (2) The atmosphere provides oxygen to breathe and an ozone layer that blocks harmful UV rays. (3) The magnetic field (from molten iron in the core) deflects cosmic rays and the solar wind, protecting the atmosphere and life.
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- Q12** It provides oxygen for respiration of almost all living things. It also forms an ozone layer that absorbs harmful UV rays, shielding living cells; and the greenhouse gases keep the temperature warm enough for liquid water.
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